M T W T F S S
Page No.:
Date:
YOUVA

5. Social Information Filtering

Q.1 What is a social media recommendation system, and how does it differ from a traditional recommendation system?

⇒

- A social media recommendation system is an algorithmic tool used on social media platforms to suggest content, people, groups or pages to users based on:-

i) User preferences and behaviors
ii) Social connections and interactions
iii) Likes, shares, comments, hashtags
iv) Real-time and historical engagement data.

- It uses social signals, trust networks and user-generated content to make personalized recommendations.
- Examples of Social Media Recommendations-

i) Facebook suggesting friends, groups, or pages.
ii) Instagram recommending posts or reels based on previous likes.
iii) Twitter recommending who to follow or trending tweets.
iv) LinkedIn recommending jobs, people or content to engage with.

Traditional Recommendation System	Social-Media based Recommendation system
1) Uses purchase history, ratings, browsing behavior and explicit preferences.	1) Social interactions, likes, comments, shares, friend's activity and engagement trends.
2) Recommendation type is products, movies, books, music, articles.	2) Recommendation type is people, posts, ads, trending topics.
3) Individual-driven	3) Network-driven.
4) Algorithm used are collaborative filtering, content-based filtering, hybrid methods.	4) Algorithm used are social graph analysis, trust-based filtering, collaborative filtering
5) Personal preferences and past interaction determine recommendation.	5) Peer influence and social engagement impact recommendations
6) Updates periodically based on browsing and purchase history.	6) Highly dynamic, update in real-time based on social trends.
7) User ratings and user-item interactions without social trust factor.	7) Uses trust score from social connections to refine suggestions.

Traditional Recommendation system	Social Media based Recommendation system
8) Usually restricted to a single platform.	8) Integrates multiple social platforms
9) For e.g., Netflix movie recommendation, Amazon product suggestions.	9) For e.g., YouTube video suggestions, Facebook friends/pages.

Page No.:
Date:

Q.2 Explain Automated, Traditional and social media recommender.

⇒

1) Automated Recommendation System

⇒ Automated systems use algorithms and data such as user behavior, ratings and preferences to provide item suggestion automatically, without human intervention.

- Features :

i) Self-improving through ML
ii) Scalable across large datasets
iii) No need for manual rule setting.

- These solutions are gaining popularity in e-commerce, advertising, entertainment and more recently training apps and platforms

- Examples :

i) Amazon recommending products based on the purchase history.
ii) Netflix recommending shows based on watched content.

2) Traditional Recommendation System

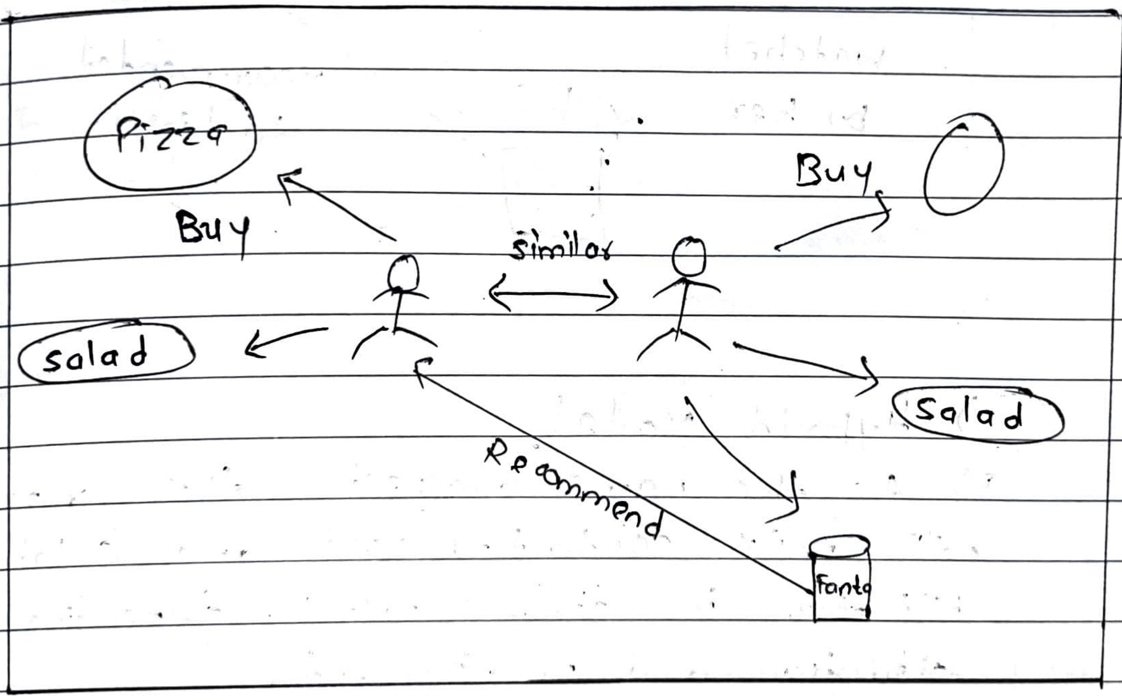
⇒ these uses historical user-item interaction data and conventional techniques to recommend items.

- Features :

i) Based on expert opinions or user reviews
ii) More static and data-dependent.

- Types :-
  - 1) Collaborative Filtering
  - 2) Content-based filtering.
  - 3) Hybrid Models.

a) Collaborative Filtering
⇒ It primarily makes recommendations based on inputs or actions from other people.

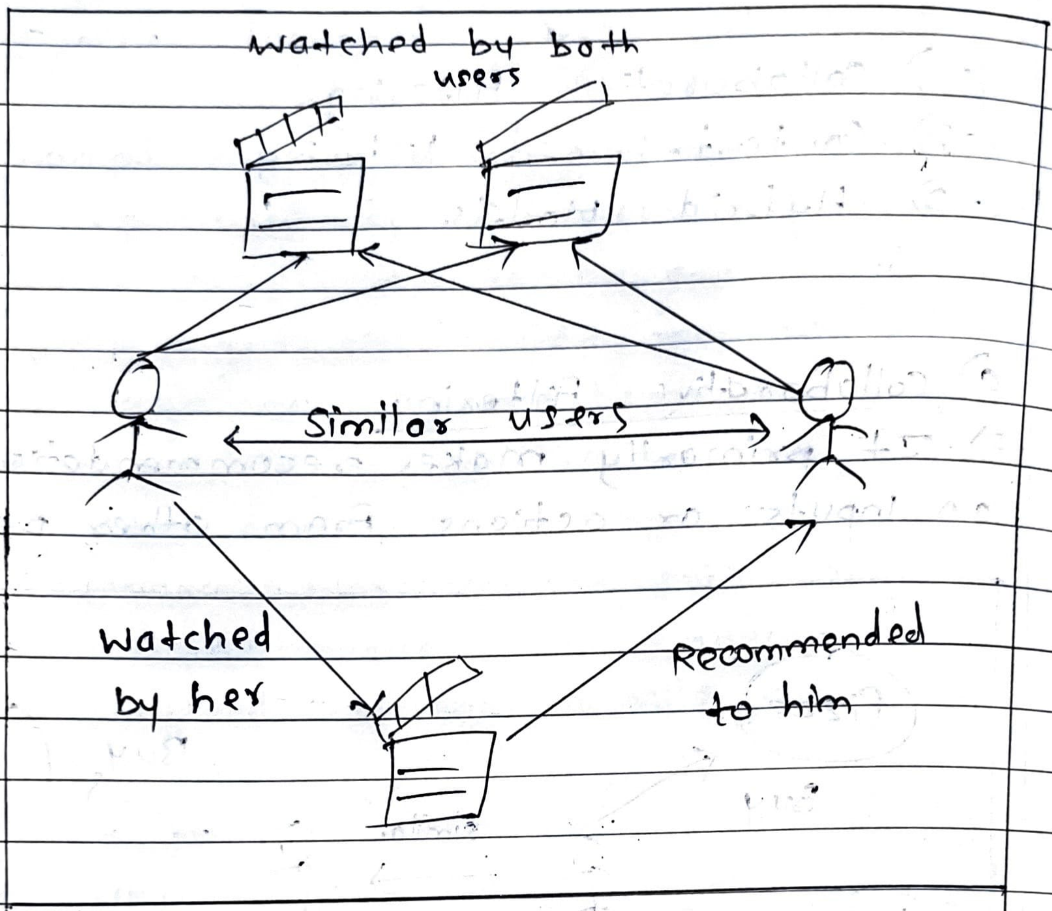


i) User-based CF : Recommends item based on what similar users have liked.
ii) Item-based CF : Suggests items similar to what a user has interacted with.

b) Content-based Filtering
⇒ Content-based system make recommendations based on user's purchase or consumption history and generally become more accurate

M T W T F S S
Page No.:
Date:
YOUVA

the more actions the user take.



c) Hybrid Model

⇒ As the name suggests, this approach will mix content-based and collaborative filtering in diverse ways, depending on the situation and requirements.

- For e.g.:
Amazon suggests products based on the browsing history and purchases made by similar users.

3) Social Media Recommender System

⇒ A social media recommendation system is an algorithmic tool used on social media platform to suggest peoples, content, groups or pages to users based on:

i) User ~~preferences~~ & behavior
ii) Social connections and interactions
iii) Likes, shares, comments, hashtags.
iv) Real-time and historical engagement data.

- For e.g.,
A Facebook suggesting events your friends are attending.

Q.3 Specify the significance of social media KPI.

⇒

- Social Media Key Performance Indicators are crucial metrics that help businesses track the success of their social media efforts.
- They align social media activities with overall business goals such as increasing brand awareness, driving website traffic and boosting conversions.
- By regularly monitoring KPIs, businesses can make data-driven decisions and maximize the impact of their marketing effort.

1) Reach

⇒ Reach measures the number of unique users who see your content, including both followers and non-followers.
- Indicates how far your content is separating across platforms.
- Helps assess brand visibility and the effectiveness of hashtags or share.

2) Engagement

⇒ Engagement tracks how users interact with your content - through likes, comments, shares and clicks.
- Reflects how well your content resonates with the audiences.
- Higher engagement often boosts visibility via platform algorithms.

3) Traffic

⇒ Traffic measures the number of users visiting your website via social media platform.

- Indicates which platform drives the most interest in your business.
- Helps identify the effectiveness of social media campaigns in leading users to your website.

4) Conversion Rate

⇒ The percentage of users who complete a desired action after clicking your social media link.

- Directly ties social media performance to tangible business outcomes.
- A high rate means your campaigns are compelling and effective.

Q.4 Explain the steps needed to formulate a social media straterogy.

⇒

- The purpose of formulating social media straterogy is to create rules and procedures to align your social media engagement with business goals.

1) Define objectives

⇒ Identify the specific goals that the business wants to achieve through social media, such as increasing brand awareness, generating leads or improving customer satisfaction.

2) Identify target audience

⇒ Determine the demographics and interests of the business's goals, audience, including age, gender, location and social media habits.

3) Research competition

⇒ Analyze the social media presence and stratergies of business's competitors to understand what is and is not working in the industry.

4) Choose social media platform

⇒ Select the social media platforms that are most relevant to the business's target audience and objectives.

M T W T F S S
Page No.:
Date:
YOUVA

5) Create a content calender
⇒ Plan and schedule the types of content that the business will post on social media, including text, images, videos and links.

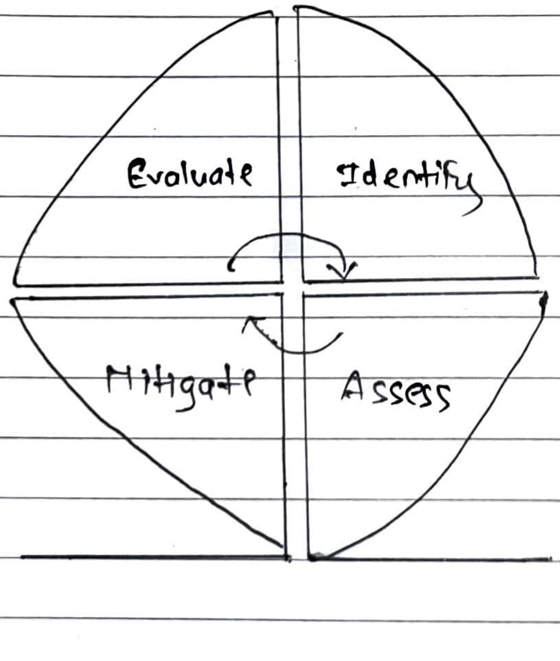
6) Engage with followers
⇒ Monitor and respond to comments and inquiries from followers and encourage user-generated content and interactions.

7) Analyze and Adjust
⇒ Use social media analytics tools to track performance of the business's social media efforts and make adjustment as needed to improve results.

Q.5 What is social media risk? Explain the four steps in social media risk management

=>

- Engaging through social media introduces new challenges related to privacy, security, data management, accessibility, social inclusion, and governance.
- Risk, in simple words, is the possibility of losing something of value such as, intellectual or physical capital.
- Social media-related risks needs to be managed properly, both from the strategic and technological points of view.
- To minimize the damage, organizations need proactive, rather than reactive, social media risk-management strategy.



1) Risk Identification

⇒ Risk identification is the process of identifying social media threats in terms of vulnerabilities and exploits that could potentially inhibit your organization from achieving its objectives.

- At this stage, your goal is to identify potential accidental or malicious risks that can come from within or outside the company.
- Examples of social-media related security breaches are hacking, information leaks, phishing and impersonation.
- Phishing and hacking are examples of malicious outsider attacks.
- The main social media risks identified were :

a) damage to reputation
b) release of confidential information
c) legal, regulatory and compilation violations
d) identity theft and hijacking
e) loss of intellectual property.

2) Risk assessment

⇒ Risk assessment is the process of assessing the probabilities and consequences of risk events if they are realized.

- The risk assessment process determines the likelihood of a social media risk event that could impact the organization economically, technically, politically and socially.

- The potential risks identified in the earlier steps are prioritized and ranked based on probability of occurrence and impact on organization.

3) Risk Mitigation

⇒ The risks prioritized and ranked in the earlier stage should be physically, technically and procedurally managed, eliminated or reduced to an acceptable level.

- Dependent on the nature of the risks, different strategies should be used
- For eg., accidental risks posed by employees can be eliminated by training, awareness programs and by having sound ~~soth~~ social media policy in place.
- Hacking attacks, for example, can be mitigated using updated antivirus systems and by creating an extra layer of security, such as two-mode authentication technique.
- Typical risk mitigation strategies are - Risk Management governance.
- New governance structures, roles, and policies should be created within your business for properly managing social media risks.

M T W T F S S
Page No.:
Date:
YOUVA

4) Risk Evaluation

⇒ Social media risk management is a continuous process.

- In the face of rapid technological, political and social change, social media risks should be periodically reviewed.
- The continuous evaluation and monitoring effort will make sure that the initial assumption made about the external and internal risks are still relevant.

Q.6 Explain the ways to measure the success of a company having social media.

=)

- The success of social media platform is measured using various key metrics that analyze user engagement, interactions and influence.
- Below are some important parameters used to assess social media performance:
  - 1) Counts
  - 2) Social sharing
  - 3) Engagement rate
  - 4) Interaction
  - 5) Referral rates

1) Counts

=> Measures the number of followers, likes, video views and shares a post receivers.

- Helps understand the reach and popularity of content and platform.

2) Social sharing

=> Tracks how often users share posts, articles or media with their followers.

- High social sharing indicates strong audience engagement and content relevance.

3) Engagement rate
⇒ Calculates the percentage of people who interact through likes and shares.
- A high engagement rate means that the content is resonating well with the audience.

4) Interaction
⇒ Measures direct Interactions such as comments, replies and discussions.
- More interactions show active audience participation and interest in the content.

5) Referral rates
⇒ Tracks how much traffic is being directed from social media to websites or other ~~group~~ platforms.
- A higher referral rates means social media is effectively driving users to external content.

Date

Q.7 Types of social media risks.

⇒
- 1) Cyberbullying
- 2) Uploading Inappropriate content
- 3) Sharing Personal Information with strangers
- 4) Privacy setting mismanagement
- 5) Phishing attacks and scams
- 6) Productivity issue
- 7) Reputational risk

1) Cyberbullying

⇒ It involves targeted harrassment, threats or insults through posts, message or comments.

- Victims may suffer anxiety, depression or a decline in self-esteem due to online abuse
- Businesses can be attacked through fake reviews, or trolling, harming their public image.

2) Uploading Inappropriate content

⇒ Sharing offensive memes or videos may lead to backlash and removal from platforms.

- Inappropriate posts can violate community guidelines, leading to account suspension.
- Content seen as insensitive can cause loss of followers, customers or brand trust.

3) Sharing personal info with strangers

⇒ Disclosing contact or location details can lead to stalking or physical threats

Page No.:
Date:
YOUVA

- Hackers may use shared info for identity theft or financial fraud.
- Oversharing makes users target for phishing, scams, and data exploitation

4) Privacy setting Mismanagement
⇒ Not updating privacy setting allows strangers to access personal posts or media.
- Public profiles expose user data to potential employees, criminals or cyberbullies
- Weak privacy controls can lead to unintended data leaks and misuse.

5) Phishing attacks and scams
⇒ Fake links or profile mimic real companies to steal login or payment details.
- Users may unknowingly install malware through malicious ads or messages
- Phishing can result in data loss, account hacking or financial theft.

6) Productivity issues
⇒ Excessive scrolling or chatting during work reduces focus and output quality.
- Notifications and distractions from social apps interfere with task completion.
- Social media addiction can lead to procrastination and burnout in professional settings.

7) Reputational Risk

⇒ Negative comments or viral ~~comments~~ complaints can damage a person's or brand's image.

- A single controversial post can spark public outrage and media ~~conver~~ coverage.
- Competitors may exploit reputation issues to gain a market or audience advantage.

Q8 Explain common social media risk-mitigation stratergies.

⇒

- Risk-mitigation stratergies for Risks :
  - 1) Cyberbullying
  - 2) Uploading Inappropriate content
  - 3) Sharing Personal info with strangers
  - 4) Privacy setting mismanagement
  - 5) Productivity issues
  - 6) Phishing Attacks and scams
  - 7) Reputation damage.

1) Cyberbullying

⇒ It involves targeted harrassment, insults or threats through posts, messages or comments.

- Mitigation stratergies :-

a) Use blocking and reporting tools provided by platform.

b) Promote digital literacy and respectful communication

c) Employ automated moderation systems to detect and filter toxic content.

2) Uploading Inappropriate content

⇒ Users may post offensive, false or sensitive content that can damage reputations or violate policies.

- Mitigation Straategies :

a) Establish and enforce clear content guidelines for users or employees.
b) Use content approval workflows for business or public accounts.
c) Leverage AI moderation tools to flag risky or explicit posts before publishing.

3) Sharing Personal Information with strangers
⇒ Revealing too much information can lead to identity theft, stalking or targeted scams.
- Mitigation straategies :-

a) Educate users on the dangers of oversharing personal info.
b) Encourage use of private profiles and strong privacy settings.

4) Misuse of Privacy settings
⇒ Not updating privacy settings allows strangers to access personal posts or media.
- Mitigation strategies :

a) Perform regular audits of account privacy settings.
b) Use alerts or reminders for users to review setting periodically.

5) Phishing attacks and scams

⇒ Users may tricked into clicking malicious links or giving away credentials through fake message.

- Mitigation strategies

a) Enable 2FA on all social media accounts.
b) Train users to identify suspicious messages and URLs.
c) Promote reporting scams to platform security teams immediately.

6) Productivity loss

⇒ Excessive social media usage can distract employees or students, reducing focus and productivity.

- Mitigation strategies :-

a) Set time limits or access controls during work or study hours.
b) Use monitoring software in organization to manage usage.

7) Reputation damage

⇒ A single negative post, review or viral complaint can harm brand reputation.

- Mitigation strategies :-

a) Monitor mentions using social listening tools
b) Respond to complaints with empathy and professionalism.

Page No.:
Date:

Q.18 Trust-based collaborative filtering is an improvement over traditional similarity-based recommendation methods. Discuss two key advantage of using trust-weighted similarity in recommendation system.

⇒

- Trust-based collaborative filtering improves traditional similarity-based methods by incorporating trust relationships between users. instead of relying solely on behavioral similarity.
- Two-key advantages :-

1) <u>Overcomes Data Sparsity & Cold start Problem</u>

⇒ Traditional similarity-based methods struggle when a users has few interactions or when dataset is sparse.

- Trust-based filtering leverages explicit trust relationships to generate better recommendations even when historical data is limited.
- For example,
  A new user on Amazon has rated very few products. Instead of relying on past behaviour a trust-based system recommends items liked by trusted users.
- On LinkedIn, job suggestions can be influenced by what trusted connections have applied for or recommended.

Page No.:

Date:

2) Improves Recommendation Accuracy & Reliability

⇒ Trust-based collaborative filtering reduces the risk of random, low-quality recommendations by prioritizing suggestions from reliable and trusted users rather than just similar users.

- For example,
  In movie recommendations, traditional systems might suggest movies based on similar viewing history, even if those users have different tastes.
- A trust-based system would prioritize movie recommendations from trusted friends or reviewers whose opinions are more meaningful.

Page No.:
Date:

Q.19 Provide an example of a real-world applications where trust scores are used to enhance recommendations. Explain how trust influences the recommendations in that scenario.

=>

1) **Amazon's Trust-based Product Recommendations:**

- Amazon incorporates trust scores by analyzing verified purchases, user reviews, and seller ratings to enhance recommendations.

i) Verified Reviews & Ratings
=> Products with high ratings from verified buyers are prioritized in recommendations.
- If a trusted reviewer rates a product highly, Amazon may suggest it to other users.

ii) Friend's Purchases & Trust Networks
=> Amazon considers what trusted users have purchased.
- For e.g., If multiple people in a trust network buy a specific brand of headphones, that brand ~~of headphones,~~ gets boosted in personalized recommendations.

3) Seller trustworthiness in Marketplace Recommendation
=> Trust scores are used to recommend products from highly rated sellers to reduce fraud.
- Amazon Marketplace show "Top Rated Sellers" who have high trust scores from previous transactions.

Page No.:
Date:

2) Airbnb :

- Airbnb also assigns trust scores based on reviews, host responses and verified profiles

i) Highly rated hosts are recommended first
⇒ Users ~~more~~ are more likely to see listings from hosts with high trust scores.
- For e.g., A host with 200+ 5star reviews will be prioritized over a new, unrated host.

ii) Guest trust score influence booking recommendation
⇒ Hosts see guest trust scores based on past stays and reviews.
- Guests with higher trust scores may get better booking options.

Page No.:

Date:

Q.20 In a collaborative filtering-based recommender system, the Pearson correlation coefficient is commonly used to measure user similarity. Explain how trust scores can enhance this method and modify recommendations.

⇒

- Collaborative filtering (CF) typically uses the Pearson correlation coefficient to measure the similarity between users based on their ratings.
- However, this approach has limitations, such as data sparsity, cold start problem and unreliable ~~things~~ ratings.
- Integreting trust scores can enhance the effectiveness of the recommendation system.
- Formula for PCC

$$PCC(A,B) = \frac{\Sigma (r_{A,i} - \bar{r}_A)(r_{B,i} - \bar{r}_B)}{\sqrt{\Sigma (r_{A,i} - \bar{r}_A)^2} \sqrt{\Sigma (r_{B,i} - \bar{r}_B)^2}}$$

where,

i) $r_{A,i}$ and $r_{B,i}$ are ratings given by users A and B to item i

ii) $\bar{r}_A$ and $\bar{r}_B$ are the average ratings of users A and B

- Key Issues in PCC:

1) Does not consider trust. PCC treats all similar users equally, even if some users are more relible or trustworthy than others

2) Cold start problem.

Page No.:

Date:

- How trust scores enhance PCC-based recommendation

- To overcome PCC's limitations, trust score can be incorporated into the similarity computation.
- Modified Trust-weighted PCC formula:

$$\text{Trust-PCC}(A,B) = \text{PCC}(A,B) \times \text{Trust}(A,B)$$

where,
Trust(A,B) represents a trust score assigned to user B by user A.

1) Trust score reduces influences from spammers, bots or unreliable users.
2) A user is more likely to trust recommendations from close friends or highly-rated reviewers.
3) Even if a new users has few ratings, they can get recommendations from trusted friends, bypassing the cold start issue.

Page No.:

Date:

F) ROI, Analysis from social media activities and campaigns.

⇒

- Return on Investment (ROI) in social media measures the profitability and effectiveness of social media activities and campaigns.
- It helps businesses determine whether their efforts on platform like Facebook, Instagram, Twitter generate measurable value.
- The basic formula for ROI is :

$$ROI = \left(\frac{\text{Net Return}}{\text{Cost of Investment}} \right) \times 100$$

Where,
Net Return = Revenue generated or value obtained from social media activities minus the cost of investment.

Cost of Investment = Total expenses incurred in running social media activities and campaigns.

- Investment includes expenses like :
  i) Social media ads
  ii) Content creation
  iii) Social media management tools.
  iv) Influencer collaborations

- Returns can be:
  i) Sales & Revenue
  ii) Lead generation

iii) Brand awareness
iv) Customer engagement

- For example;
A brand spends \$ 1,000 on Instagram ads and earns \$ 3,000 in sales.
The ROI could be:

$$\frac{3000 - 1000}{1000} \times 100 = \frac{2000}{1000} \times 100$$
$$= 200\%$$

g) Business use of social media
⇒

- Social media has become a powerful tool for businesses to connect with customers, promote products and build brand loyalty.

1) Brand Awareness & Marketing
⇒ Reach a global audience with minimal cost.
- Run targeted ad campaign to specific demographics.
- Increase visibility through viral content and influencer marketing.
- For eg:, A fashion brand uses Instagram reels to showcase new collection and attract young audiences.

Page No.:
Date:

2) Customer engagement
⇒ Engage with customers through ~~custome~~ comments, DMs, and live sessions.
- Build brand loyalty by responding to customer queries quickly.
- For e.g., Starbucks uses Twitter for customer engagement and promotions, responding to tweets and sharing user-generated content

3) Sales & Lead generation
⇒ Drive website traffic through social media ads.
- Use social commerce. Collect leads via LinkedIn and Facebook lead-generation forms.

4) Customer support & Feedback collection
⇒ Provide 24/7 customer service through chatbots and DMs.
- Monitor brand sentiment through social listening tools.
- For e.g., A telecom company like Verizon uses Twitter for real-time customer service, answering queries and resolving complaints.

Page No.:
Date:

Q. 21
=>

Soln :-

i) Alice and Bob
=>

$$r = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2}}$$

x (Alice)	y (Bob)	$(x-\bar{x})$	$(x-\bar{x})^2$	$(y-\bar{y})$	$(y-\bar{y})^2$
5	3	1.8	3.24	-0.2	0.04
4	5	0.8	0.64	1.8	3.24
3	2	-0.2	0.04	-1.2	1.44
3	5	-0.2	0.04	1.8	3.24
1	1	-2.2	4.84	-2.2	4.84
16	16		8.8		12.8

$$\bar{x} = \frac{5+4+3+3+1}{5} = 3.2$$

$$\bar{y} = \frac{3+5+2+5+1}{5} = 3.2$$

$$r = \frac{[(1.8 \times (-0.2)) + (0.8 \times 1.8) + (-0.2 \times -1.2) + (-0.2 \times 1.8) + (-2.2 \times -2.2)]}{\sqrt{8.8 \times 12.8}}$$

$$= \frac{-0.36 + 1.440 + 0.24 - 0.36 + 4.84}{\sqrt{112.64}}$$

$$= 0.5465$$

Page No.:

Date:

$$Result_{Bob} = 0.5465 \times 0.6$$
$$= 0.3279$$

ii) Alice and chuck

$\Rightarrow$

x	y	$(x-\bar{x})$	$(x-\bar{x})^2$	$(y-\bar{y})$	$(y-\bar{y})^2$
5	4	1.8	3.24	1.4	1.96
4	3	0.8	0.64	0.4	0.16
3	2	-0.2	0.04	-0.6	0.36
3	2	-0.2	0.04	-0.6	0.36
1	2	-2.2	4.84	-0.6	0.36
16	13		8.8		3.2

$\bar{x} = 3.2$

$\bar{y} = 2.6$

$$r = \frac{[(1.8 \times 1.4) + (0.8 \times 0.4) + (-0.2 \times -0.6) + (-0.2 \times -0.6) + (-0.2 \times -0.6)]}{\sqrt{8.8 \times 3.2}}$$

$$= \frac{1.76}{5.31}$$

$$= 0.3315$$

$$Result_{chuck} = 0.3315 \times 0.9$$
$$= 0.2984$$

- Bob will have greater influence on Alice's recommendation.

Q.11 What are possible social media risks?
=>
- Social media gives brand the tools to easily communicate with a global audience, but there are a few ways things can go wrong leading to unauthorized based communication:

1) User authorization
=> User authorization refers to what a user is able to do on a platform, especially a permission level that dictates if someone can view or make changes.
- On social media, it's important to make sure each individual with account access has the right authorization level.

2) Phishing
=> Phising occurs when a malicious third-party attempts to impersonate a brand and communicate with customers to get them to reveal sensitive personal information.
- On social media, some individuals will create fake brand accounts and publish posts designed to look like promotional offers.

Page No.:
Date:

using the same language used by the actual organization.
- By clicking on the link, the consumer may enable to impersonator to access sensitive information such as passwords and credit card numbers.

3) Non-secured password
⇒ LMG security notes that an 8-character password can be cracked in under 8 hours, a 10-character password takes 8 years and 12-character password takes 77,000 years.
- The firm notes that organizations often opt for efficiency over security, choosing short and easy-to-remember passwords which make them vulnerable to hackers

4) Reputation Damage
⇒ Negative reviews, misinformation, or viral backlash can harm a brand's image.
- For e.g., A controversial post by a company goes viral, leading to public outrage and boycotts.

5) Location setting
⇒ Location app setting may still track user whereabouts. Even if someone turns off their location setting, there are other ways to target a device's location.
- The use of public wifi, cellphone towers and websites can also track user locations.